Application of digital signal processor to snore cancellation

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***Abstract*—This paper presents the application of a digital signal processor (DSP) to snore cancellation. The algorithm, which includes a finite impulse response (FIR) filter and the least mean square (LMS) method to tune the weights in filter, is adopted to design an anti-snore signal to destructively interfere the snore. Experimental results show the effectiveness of proposed method is up to 5-10dB.**

***Keywords*—DSP, snore cancellation, FIR, LMS**

I. INTRODUCTION

Given the performance to apply artificial anti-noise to destructively interfere the undesired noise, the technique of active noise control has been widely investigated by using a high speed digital signal processor. This method, also called active noise control (ANC), generates the anti-noise interference with the same magnitude and a 180 degree phase shift to the unwanted noise to disrupt it [1-4]. ANC system is general well-organized to reduce low frequency noise. Besides, the passive noise control (PNC) is only efficient in canceling out high frequency noise by using sound-absorbing material. But most annoying noise has its major power in a low frequency; it is the reason why ANC has been widely investigated [5-7].

This paper discusses the cancellation of a low frequency snore signal by using ANC based method in a DSP. Generally speaking, snoring is a high up problem in nowadays. The volume of severe snoring can reach up to 90 dB, which not only cause hearing loss but also disturb the sleep of bed partner. It also linked to extremely sleepiness in daytime of snorer and his bed partner, leading to the loss of productivity in the work, safety of driving and occupation. Most solutions to this problem use passive earmuffs or earplug for the bed partner. But these facilities are uncomfortable to be wear when sleep. Another method like anti-snore pillow or tooth socket for the snorer is uncomfortable, either. A period of snore can be divided into two components, inspiration and expiration. The frequency of snore is also in low frequency, between 100 to 300 Hz. Accordingly, based on the theory of superposition, the ANC system is a better way to reduce the snore noise by generating an artificial anti-snore signal [13].

This paper uses the popular filtered-X least mean square (FXLMS) method in ANC to derive the proper signal to cancel the undesired snore noise. By using a DSP chip, this method can be realized through the estimation of secondary path, from the location of secondary speaker to the error microphone, and then tune the weights of the FIR filter to generate the anti-snore noise. This artificial noise has the same magnitude and 180 degrees phase shift of the annoying snore. Therefore, the anti-snore noise can interfere the snore signal, producing a silent zone at the ears of bed partner.

II. FXLMS ALGORITHM

Utilizing FXLMS algorithm, the conventional ANC system is shown in the Fig. 1 [8-12]. The plant model $H_p(z)$ represents the primary speaker and the model $P(z)$ denotes the mathematical transfer function of the acoustic duct, the input signal $x(k)$ represents the kth unwanted noise sample and the residual noise $e(k)$ is measured by an error microphone with plant model $M_e(z)$. The output signal of the FXLMS $u(k)$ drives an anti-noise speaker with the transfer function $H_s(z)$ to cancel the unwanted noise in the duct. The acoustic transfer function between the location of the anti-noise speaker and the error microphone is referred to as the error path which is represented by $H_e(z)$. The Nth-order ANC with weighting parameters $W(k)$ is updated by the FXLMS algorithm. Thus, the residual error noise $e(k)$ can be expressed as

$$e(k) = x(k)H_p(z)P(z)H_e(z)M_e(z) + y(k)H_s(z)H_e(z)M_e(z) \tag{1}$$

where $y(k) = W(k)^T X(k)$, the weighting parameter $W(k) = [w_0(k), w_1(k), ..., w_{N-1}(k)]^T$ and the input noise $X(k) = [x(k), x(k-1), ..., x(k-N+1)]^T$. The corresponding gradient estimation and adaptations of the weights are taken as:

$$\nabla_k = \frac{\partial e^2(k)}{\partial W(k)} = 2e(k)[X(k)H_s(z)H_e(z)M_e(z)] \tag{2}$$

$$W(k+1) = W(k) - 2\mu e(k)[X(k)H_s(z)H_e(z)M_e(z)] \tag{3}$$

where μ is the learning rate. Equation (3) is called FXLMS algorithm, which differs from the conventional LMS algorithm (4) at the correction term,

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$$W(k+1) = W(k) - 2\mu e(k)X(k) \quad (4)$$

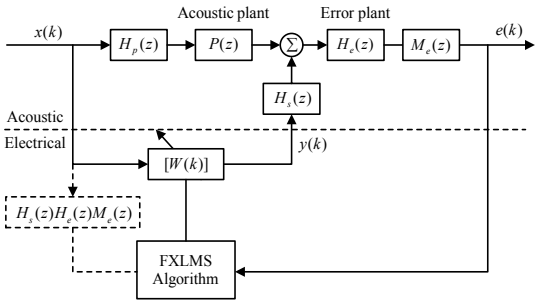


Fig. 1. Conventional ANC system with FXLMS algorithm.

The block diagram of using FXLMS in a snore cancellation system can be described as in Fig. 2. In Fig. 2, we use two speaker, mounted on a board, to be the secondary speakers of left and right ears. Two microphones are placed in front of the speakers to measure the residual noise. Another speaker, located at the left side of the board, plays the snore continuously.

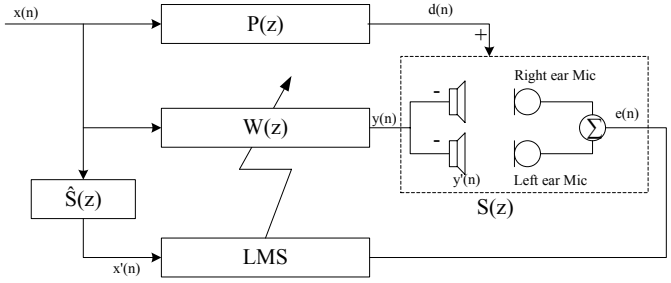


Fig. 2. Block diagram of snore cancellation system using FXLMS algorithm.

A reference microphone and two error microphones measure the snore and residual snore for the use of ANC. In this diagram, the secondary path is from the location of secondary speaker to error microphone, denoted by $S(z)$. One has to identify it by measuring the frequency response between secondary speaker and secondary speaker and microphone. In order to obtain this information, one sent White noise to the secondary speaker and used error microphone to measure. We derived the frequency response $S(z)$ by a digital signal analyzer - Agilent 35670A. Accordingly, we can use this information to derive the right correction term of LMS algorithm to tune the coefficients of FIR filter by a DSP – TMS320C6713.

So, we have the similar results such as those in Fig. 1.

$$e(n) = x(n)P(z)S(z) + y(n)S(z) \quad (5)$$

where $y(n) = W(n)^T X(n)$, the weighting parameter $W(n) = [w_0(n), w_1(n), ..., w_{N-1}(n)]^T$ and the input noise $X(n) = [x(n), x(n-1), ..., x(n-N+1)]^T$. The corresponding gradient estimation and adaptations of the weights are taken as:

$$\nabla_k = \frac{\partial e^2(n)}{\partial W(n)} = 2e(n)[X(n)S(z)] \quad (6)$$

$$W(n+1) = W(n) - 2\mu e(n)[X(n)S(z)] \quad (7)$$

where μ is the learning rate. Equation (7) is the FXLMS algorithm in a snore cancellation system, which also differs from the conventional LMS algorithm (4) at the correction term.

III. Experimental Results

A. Error Microphone In Front Of Speaker

The overall experimental setup is shown in Fig. 3. One uses a microprocessor of Texas Instrument (TI-TMS320C6713) to be the core of ANC system. A Harman Kardon soundsticks – II speaker, which has its special ability to play signal at low frequency, is applied to show the snorer. A 100cm X 40cm board is used to be the headboard in bed. Two secondary speakers, with diameter 8 inches, are mounted on the board to play the anti-snore signal. Three free field microphones, with 1/2 inches diameters, were made by PCB Piezotronics Co. Ltd. A Marantz PM6100SA power amplifier is used to be the driver of secondary speaker, and an ICP power is used to provide the power to microphone. Agilent - digital spectrum analyzer 35670A is used to measure the information of noise in ANC system.

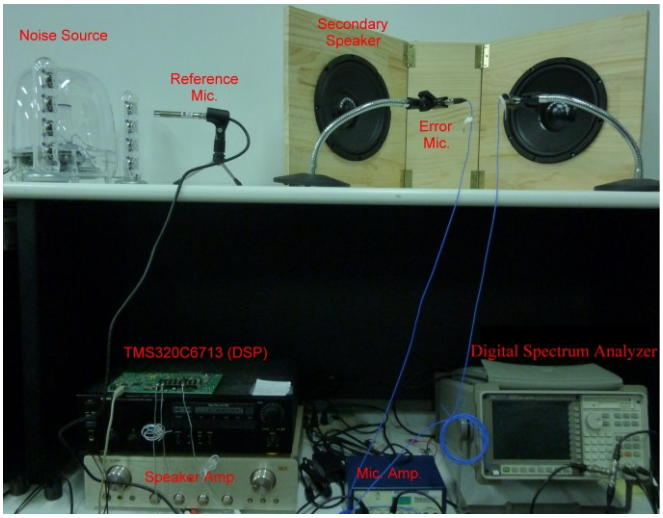


Fig. 3. Experimental setup of ANC system for snore cancellation.

Because the snore is a fast changing noise with the frequency range 100-300Hz, one applied several kinds of noises to verify the performance of the proposed system. In the first two experiments, the undesired noises were supposed to be narrowband noises. One used 200Hz and 300Hz to be the

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undesired signals, and the results after ANC activation were shown in Figs. 4 and 5.

In Fig. 4, One can find the undesired 200 Hz noise is up to -45dB, while the result is -90dB when ANC is on, shown by the blue and red lines. The performance is obvious.

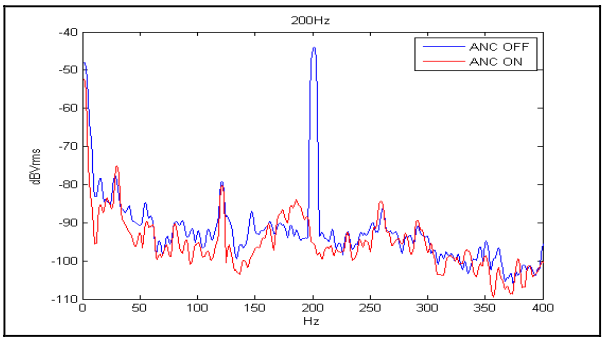


Fig. 4. Narrowband noise with 200Hz, blue: ANC OFF and red, ANC ON.

The frequency of undesired noise changed to 300Hz in the following experiment, but the proposed ANC system still can cancel the noise very well, from -50dB to -95dB. The performance is almost 45dB, shown in Fig. 5.

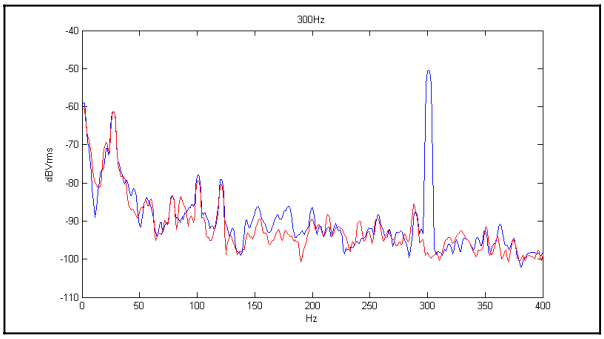


Fig. 5. Narrowband noise with 300Hz, blue: ANC OFF and red, ANC ON.

Next experiment shows the performance of reducing broadband noise. We use the random noise between 200Hz to 250Hz to be the undesired noise. In Fig. 6, we can find the performance of proposed algorithm is still good. The performance is about 10dB. The increasing noise at about 350Hz is due to the Waterbed effect, which is still under tolerance.

Because the snorer is usually located at the range of 200-300Hz, we show the performance of proposed method to cancel 250-300Hz broadband noise. The result is shown in Fig. 7. The BLUE and RED lines plot the broadband noises of ANC OFF and ON, respectively. One can find the performance still works well. The performance is about 10-15dB.

By using narrowband and broadband noises within the frequency ranges 200Hz to 300Hz, one can find the proposed FXLMS algorithm can cancel the undesired noise in the open space of bedroom.

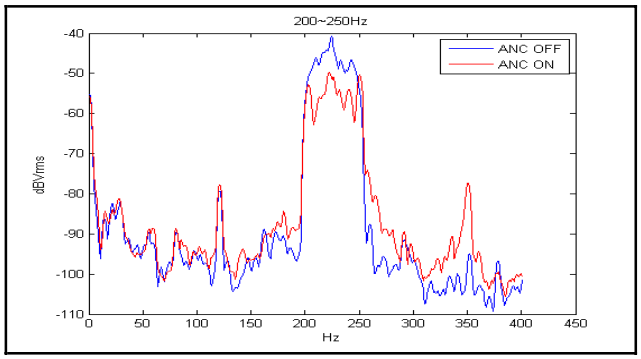


Fig. 6. Broadband noise between 200-250Hz, blue: ANC OFF and red, ANC ON.

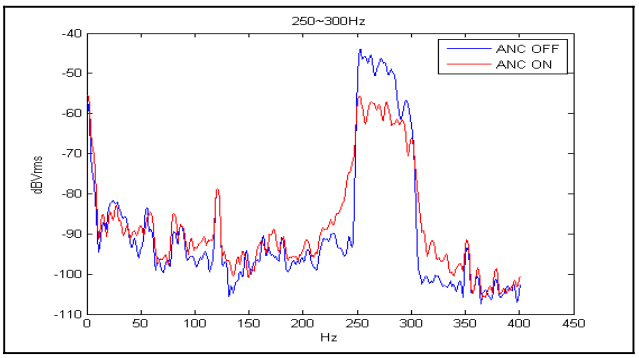
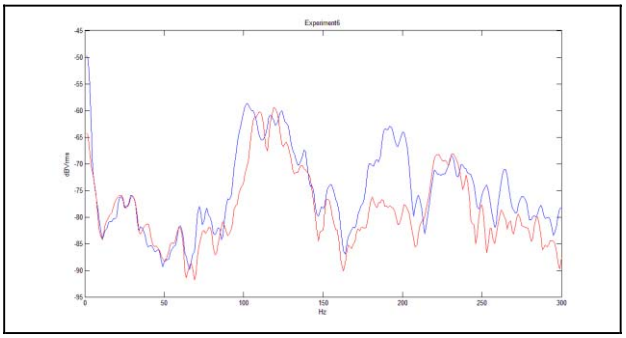


Fig. 7. Broadband noise between 250-300Hz, blue: ANC OFF and red, ANC ON.

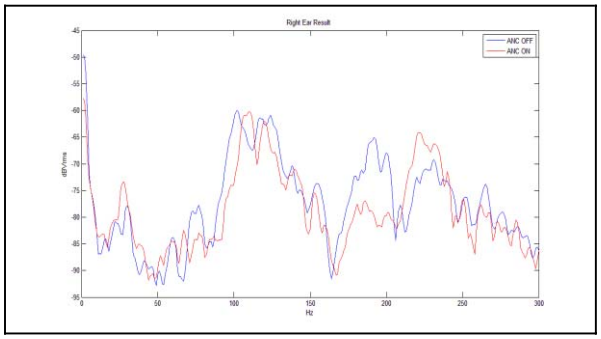
Last example examines the performance of proposed method to cancel the undesired snore signal. One uses a snore signal, obtained from the Internet, being played by the Harman Kardon soundsticks – II to be the noise source. The two error microphones, in front of the secondary speakers, can be thought as the positions of left and right ears of roommate beside the snorer. One also applies the microphones to imitate the residual snore after activation of the ANC. In Fig. 8, one averages the snore signal and plot its power spectrum by the BLUE ones. After the ANC is ON, the result is shown by the RED lines. First of all, one can find the main power snore is located at 100-300Hz. In Fig. 8(a), the canceling performance of left ear is obvious at the frequency range 170-220Hz. The enhancement is about 15dB.

We next examine the results at the right side. The ANC still works well between 170-220Hz at the right ear. The performance is also about 15dB. However, the frequency range between 100-150Hz does not work by ANC. Noise at this range can not be restrained by the proposed method. It could be the problem of the secondary speaker performs badly at the lower frequency. Besides, the lower frequency noise is also insensitive to human ears.

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(a)

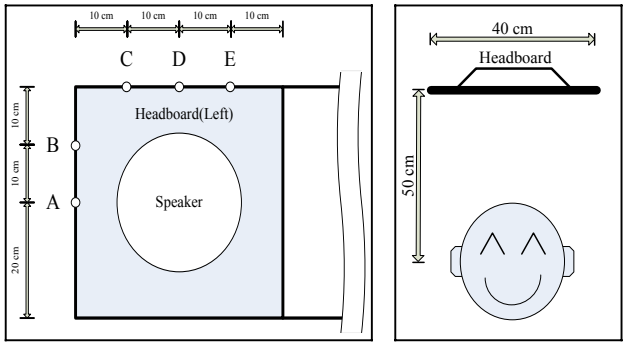


(b)

Fig. 8. The results of ANC to cancel the snore signal, blue: ANC OFF and red, ANC ON, (a) left ear (b) right ear.

B. Error Microphone mounted on Headboard

To mount the error microphone on the people's ear is not the good idea. So we choose five different positions to mount error microphone on the headboard. The five error microphone positions are shown in the Fig. 9(A). And the experiments set up is shown in Fig. 9(B). Mount the error microphone on the headboard and start the ANC system to measure the residual noise near the people's ear.



(A) (B)

Fig. 9. Experimental Setup (A) Side view (B) Vertical view

The experiments results are shown in Fig. 10. The result showing the best position of the error microphone that mounted on the headboard.

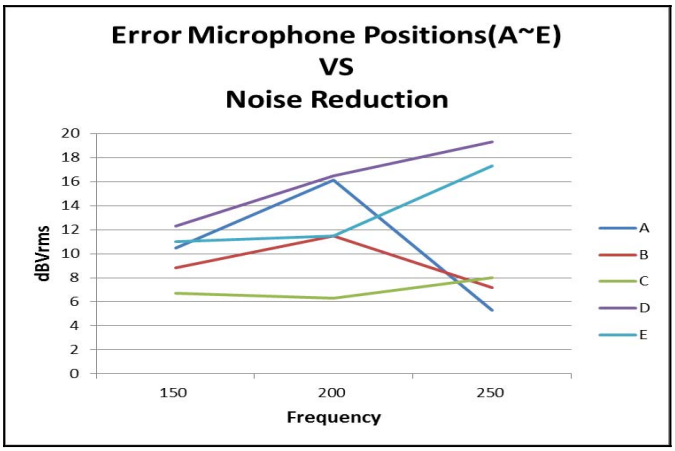
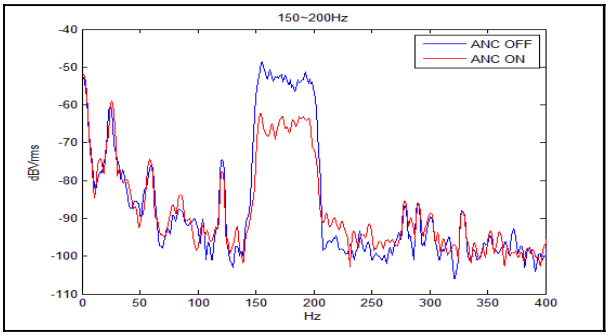
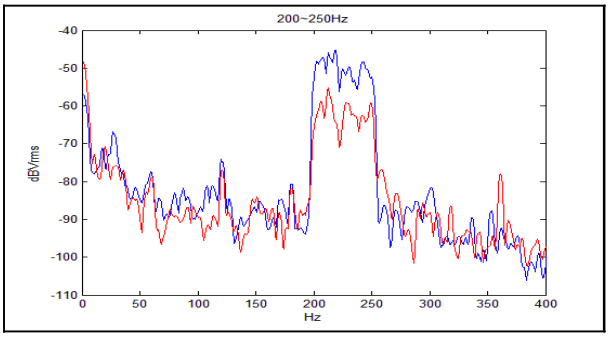


Fig. 10.

From Fig. 11, it is show the result that the residual noise nears the people's ear. For single frequency noise the reduction have 12~20dBvrms. For broadband noise the reduction have 10dBvrms in average.



(A)



(B)

Fig. 11. Broadband noise between 150~200Hz and 200~250Hz

IV. CONCLUSIONS

This paper discusses using the properties of ANC system in a snore cancellation problem. Experimental results are verified through the implementation of ANC system by a DSP chip. The results prove that the snore signal can be restrained up to 5-15 dB. A silent zone is also created to improve the sleep quality of the roommate beside a snorer.

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